

Cryptology Assignment 3

1. Determine multiplicative inverse module 26 for each of the following integers.

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25 mod 26

→ i) multiplicative inverse of 1

$$26 = 1(26) + 0$$

$$\Rightarrow 1(1) = 1$$

$$1^{-1} \equiv 1 \pmod{26}$$

∴ multiplicative inverse of 1 = 1

x ii) multiplicative inverse of 5.

$$26 = 5(5) + 1$$

$$1 = 26 - 5(5)$$

$$1 = 5(5) + 26(-1)$$

$$-5 + 26 = 21$$

$$5^{-1} \equiv 21 \pmod{26}$$

∴ multiplicative inverse of 5 = 21

iii) multiplicative inverse of 3

$$26 = 3(8) + 2$$

$$2 = 26 - 3(8)$$

$$1 = 3 - 2 \rightarrow \textcircled{0}$$

$$2 = 26 - 3(8)$$

$$1 = 3 - (26 - 3(8))$$

$$1 = 3 - 26 + 3(8)$$

$$1 = 3(9) + 26(-1)$$

$$3^{-1} \equiv 9 \pmod{26}$$

∴ multiplicative inverse of 3 = 9

iv) multiplicative inverse of 7

$$26 = 7(3) + 5, \quad 5 = 26 - 7(3)$$

$$7 = 5(1) + 2, \quad 2 = 7 - 5 \rightarrow \textcircled{0}$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2) \rightarrow \textcircled{0}$$

$$1 = 5 - 2(7 - 5), \text{ sub } \textcircled{0} \text{ in } \textcircled{0}$$

$$1 = 5 - 2 \times 7 + 2 \times 5$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$1 = 3 \times 26 - 3 \times (7(3)) - 2(7)$$

$$1 = 8 \times 26 - 11 \times 7$$

$$1 = (-11) \times 7 + 8 \times 26, \quad -11 + 26 = 15$$

$$7^{-1} \equiv 15 \pmod{26}$$

∴ multiplicative inverse of 7 = 15.

v) multiplicative inverse of 9

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8, \quad 8 = 26 - 9(2)$$

$$1 = 9 - (26 - 9(2))$$

$$1 = 9 + 9(2) - 26$$

$$1 = 3(9) - 26$$

$$9^{-1} \equiv 3 \pmod{26}$$

$\therefore$ multiplicative inverse of 9 is 3.

vi) multiplicative inverse of 11;

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3, \quad 3 = 11 - 4(2)$$

$$4 = 3(1) + 1, \quad 4 = 26 - 11(2)$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2)) = 4 - 11 + 4(2)$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2)) - 11$$

$$1 = 3 \times 26 - 3 \times (11(2)) - 11$$

$$1 = 3 \times 26 - 7 \times 11$$

$$1 = (-7)11 + 3 \times 26$$

$$11^{-1} \equiv 19 \pmod{26}$$

$\therefore$ multiplicative inverse of 11 is 19.

vii) multiplicative inverse of 15,

$$26 = 15(1) + 11, \quad 11 = 26 - 15$$

$$15 = 11(1) + 4, \quad 4 = 15 - 11(1)$$

$$11 = 4(2) + 3, \quad 3 = 11 - 4(2)$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2))$$

$$1 = 3(4) - 11$$

$$1 = 3(15 - 11(1)) - 11$$

$$1 = 3 \times 15 - 4 \times 11$$

$$1 = 3 \times 15 - 4(26 - 15)$$

$$1 = 7(15) - 4(26)$$

$$15^{-1} \equiv 7 \pmod{26}$$

$\therefore$ multiplicative inverse of 15 is 7.

viii) multiplicative inverse of 17,

$$26 = 17(1) + 9, \quad 9 = 26 - 17(1)$$

$$17 = 9(1) + 8, \quad 8 = 17 - 9(1)$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8$$

$$1 = 9 - (17 - 9(1))$$

$$1 = 2(9) - 17$$

$$1 = 2(26 - 17(1)) - 17$$

$$1 = 2 \times 26 - 2 \times 17(1) - 17$$

$$1 = 2 \times 26 - 13 \times (17)$$

$$1 = (-3)17 - 2(26), \quad 17^{-1} \equiv 23 \pmod{26}$$

∴ multiplicative inverse

of 17 is 23

ix) multiplicative inverse of 19,

$$26 = 19(1) + 7, \quad 7 = 26 - 19$$

$$19 = 7(2) + 5, \quad 5 = 19 - 7(2)$$

$$7 = 5(1) + 2, \quad 2 = 7 - 5(1)$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 5 - 2 \times 7 + 2(5(1))$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(19 - 7(2)) - 2(7)$$

$$1 = 3 \times 19 - 8(7)$$

$$1 = 3(19) - 8(26 - 19)$$

$$1 = 11(19) - 8(26)$$

$$19^{-1} \equiv 11 \pmod{26}$$

∴ multiplicative inverse

of 19 is 11.

x) inverse of 21,

$$26 = 21(1) + 5$$

$$21 = 5(4) + 1$$

$$1 = 21 - 5(4)$$

$$1 = 21 - 4(26 - 21)$$

$$1 = 21 - 4(26) + 4(21)$$

$$1 = 5(21) - 4(26)$$

$$21^{-1} \equiv 5 \pmod{26}$$

multiplicative inverse of 21 is 5.

inverse of 23

$$26 = 23(1) + 3 \quad , \quad 3 = 26 - 23$$

$$23 = 3(7) + 2 \quad , \quad 2 = 23 - 3(7)$$

$$3 = 2(1) + 1$$

$$1 = 3 - 2(1)$$

$$1 = 3 - (23 - 3(7))$$

$$1 = 8(3) - 23$$

$$1 = 8(26 - 23) - 23$$

$$1 = (-9)23 + 208(8)$$

$$26 - 9 = 17$$

$$23^{-1} = 17 \pmod{26}$$

multiplicative inverse of

23 is 17

inverse of 25

$$26 = 25(1) + 1$$

$$1 = 26 - 25$$

$$1 = (-1)25 + 26$$

$$26 - 1 = 25$$

$$25^{-1} = 25 \pmod{26}$$

multiplicative inverse

of 25 is 25.

2. Choose any string of your choice to encrypt using the affine cipher with key value multipliers and additive.

→ for encryption,

$$P = (K_1 \cdot P + K_2) \pmod{26}$$

given $K_1 = 5$, $K_2 = 8$

P	T	P	$K_1 \times P$	$K_1 \times P + K_2$	$\pmod{26}$	
G	6	$5 \times 6 = 30$	$30 + 8$	38	12	m
O	14	$5 \times 14 = 70$	$70 + 8$	78	0	a
O	14	$5 \times 14 = 70$	$70 + 8$	78	0	a
D	3	$5 \times 3 = 15$	$15 + 8$	23	23	x
L	11	$5 \times 11 = 55$	$55 + 8$	63	11	L
U	20	$5 \times 20 = 100$	$100 + 8$	108	4	e
L	2	$5 \times 2 = 10$	$10 + 8$	18	18	s
K	10	$5 \times 10 = 50$	$50 + 8$	58	6	g

ciphertext = MAAxLESg

3. Encrypt 'HI' using the affine cipher with $a=11$ and $b=5$.

→ for encryption: $C = E(K_1 * P + K_2) \text{ mod } 26$

$$K_1 = 11, K_2 = 5$$

plaintext		$K_1 * P$		$+ K_2$	$\text{mod } 26$	cipher
H	7	$11 \times 7 = 77$		$77 + 5 = 82$	4	E
I	8	$11 \times 8 = 88$		$88 + 5 = 93$	15	P

cipher text: EP

4. the ciphertext 'EP' was encrypted using affine function $ax + b \text{ mod } 26$, find plaintext

→ for decryption, $P = K^{-1}(C - K_2) \text{ mod } 26$

$$K_1 = 11, K_2 = 5$$

$$11^{-1} \text{ mod } 26$$

$$26 = 11(2) + 4, 4 = 26 - 11(2)$$

$$4 = 11(-2) + 1$$

$$1 = 11 - 4$$

$$1 = 11 - (26 - 11(2))$$

$$1 = 3(11) - 26$$

$$11^{-1} \text{ mod } 26 = 3$$

ciphertext		$K^{-1}(C - K_2)$	$\text{mod } 26$	PT
E	4	$3(4 - 5)$	54	C
P	15	$3(15 - 5)$	30	A
		$3(17 - 5)$	45	T

Plaintext = CAT

5. decrypt the cipher text using affine cipher, key x t u w o b x z y using the key 3 (mod) 9 4

→ for decryption, $P = K^{-1}(C - K_2) \text{ mod } 26$

cipher text: K O Y X T U W O B X Z Y

$$K_1 = 3, K_2 = 4$$

$$3^{-1} \text{ mod } 26$$

$$26 = 3(8) + 2, 2 = 26 - 3(8)$$

$$3 = 2(13) + 1$$

$$1 = 3 - 2$$

$$1 = 3 - (26 - 3(8))$$

$$1 = 3(9) - 26$$

$$3^{-1} \bmod 26 = 9$$

C/T		$K^{-1} (C - K_0) \bmod 26$	$\bmod 26$	P/T
K	10	$9(10 - 4) = 54$	2	C
D	3	$9(3 - 4) = -9$	17	R
Y	24	$9(24 - 4) = 180$	24	Y
X	23	$9(23 - 4) = 171$	15	P
J	9	$9(9 - 4) = 45$	19	T
U	20	$9(20 - 4) = 144$	14	O
W	22	$9(22 - 4) = 162$	6	G
D	3	$9(3 - 4) = -9$	17	R
E	4	$9(4 - 4) = 0$	0	A
X	23	$9(23 - 4) = 171$	15	P
Z	25	$9(25 - 4) = 189$	7	H
Y	24	$9(24 - 4) = 180$	4	Y

plaintext: CRYPTOGRAPHY

6. we $K_1 = 13$ (multiplier), $K_2 = 4$ (even though it is not a valid key try to encrypt)

Encrypt the string "input" & "alter" using the above key & find what the issue is.

→ for encryption, $C = E(K_1 * P + K_2) \bmod 26$

$$K_1 = 13, K_2 = 4$$

P/T		$K_1 * P$	$+ K_2$	
I	8		$104 + 4 = 108$	4
N	13	$13 \times 8 = 104$	$169 + 4 = 173$	17
P	15	$13 \times 13 = 169$	$195 + 4 = 199$	17
U	20	$13 \times 15 = 195$	$260 + 4 = 264$	4
T	19	$13 \times 20 = 260$	$247 + 4 = 251$	17
A	0	$13 \times 19 = 247$	$0 + 4 = 4$	4
L	11	$13 \times 0 = 0$	$142 + 4 = 147$	17
T	19	$13 \times 11 = 143$	$247 + 4 = 251$	17
E	4	$13 \times 19 = 247$	$52 + 4 = 56$	4
R	17	$13 \times 4 = 52$	$221 + 4 = 225$	17

● $k_1 = 13$ is not prime with 26, the entire 26 letter alphabet is compressed into just two apparent letters (E & R)

7. Find the inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \text{ mod } 26$

$$\rightarrow \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} = (3 \times 7) - (5 \times 2) = 21 - 10 = 11$$

$$11^{-1} \text{ mod } 26 = 26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$11 = 3(1) + 1$$

$$1 = 4 - 3(1)$$

$$1 = 4 - 3(11 - 4(2))$$

$$1 = (4) - (11)$$

$$1 = (26 - 11(2)) - (11) = -6(11) - (11) + 3 \times 26$$

$$1 = (-7)11 + 3 \times 26, -7 + 26 = 19$$

$$11^{-1} \text{ mod } 26 = 19$$

$$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 133 & -38 \\ -45 & 57 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \rightarrow K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8. Find the inverse of the following matrix whose entries are considered modulo 26:

$$\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} = (11 \times 3) - (13 \times 2) = (33 - 26) = 7$$

$$7^{-1} \text{ mod } 26 \rightarrow 26 = 7(3) + 5 \rightarrow 1 = 5 - 2(2)$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5)$$

$$1 = 3(5) - 2(7)$$

$$\rightarrow 1 = 3(26 - 7(3)) - 2(7)$$

$$1 = (-11)7 + 3(26) \Rightarrow -11 + 26 = 15 \rightarrow 7^{-1} \text{ mod } 26 = 15$$

$$15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} = \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9. is the key valid? $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

$$\rightarrow \begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix} = (2 \times 21) - (13 \times 14) = (42) - (182) = -140$$

$\rightarrow -140$ is not coprime with 26.

10. Encrypt using 2×2 Hill cipher for the plaintext "SOLVED" using
 $\begin{matrix} 3 & 8 \\ 7 & 11 \end{matrix}$

⇒ for encryption $C = P \cdot K \pmod{26}$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = (3 \times 11) - (7 \times 8) = (33 - 56) = 23$$

23 is non-zero

23 is co-prime with 26

plaintext: SOLVED

$$i) P = \begin{bmatrix} S \\ O \end{bmatrix} = \begin{bmatrix} 18 \\ 14 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \cdot \begin{bmatrix} 18 \\ 14 \end{bmatrix} = \begin{bmatrix} 3 \times 18 + 8 \times 14 \\ 7 \times 18 + 11 \times 14 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} K \\ U \end{bmatrix}$$

$$ii) P = \begin{bmatrix} L \\ V \end{bmatrix} = \begin{bmatrix} 11 \\ 21 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} = \begin{bmatrix} 3 \times 11 + 8 \times 21 \\ 7 \times 11 + 11 \times 21 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 22 \end{bmatrix} = \begin{bmatrix} T \\ W \end{bmatrix}$$

$$iii) P = \begin{bmatrix} E \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \times 4 + 8 \times 3 \\ 7 \times 4 + 11 \times 3 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 \\ 9 \end{bmatrix} = \begin{bmatrix} K \\ J \end{bmatrix}$$

Ciphertext: KUTWKJ